Generative AI Fundamentals

Practice Topics

What is Artificial Intelligence?

Artificial Intelligence (AI) is the process of making computer enable to perform tasks that normally require human intelligence. These tasks include understanding language, recognizing images ec.

What is Machine Learning?

Machine Learning (ML) is a branch of AI where computers learn patterns from data without being explicitly programmed. After training on examples, an ML model can make predictions on new data.

What is Deep Learning?

Deep Learning is a type of machine learning that uses neural networks with many layers. It is especially good at handling complex data like images and speech text because it automatically learns useful features.

What is Generative AI?

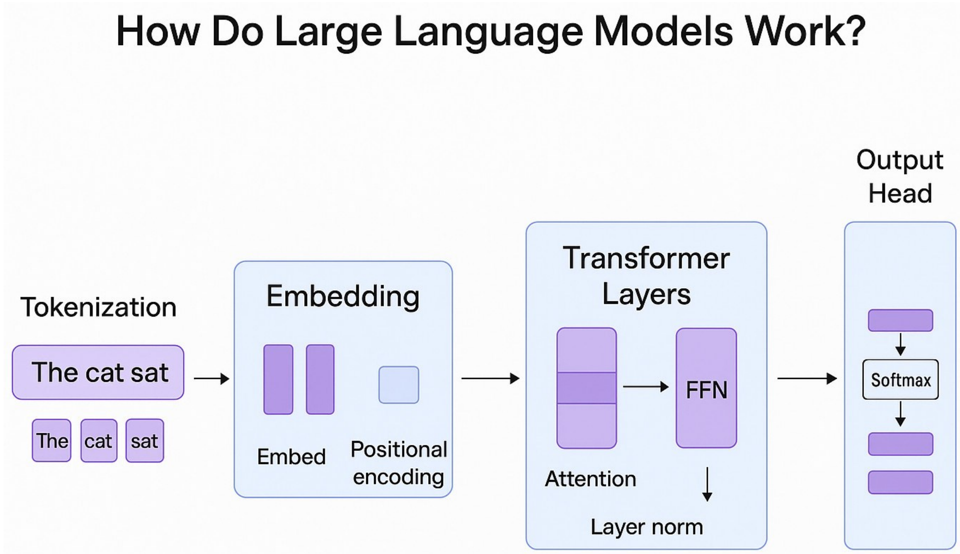
Generative AI is a type of AI that creates new content such as text, images, music, code, or videos based on what it learned during training.

What is a Large Language Model (LLM)?

A Large Language Model (LLM) is an AI model trained on large amounts of text. It learns language patterns so it can answer questions, write content, summarize information, translate languages, and help in coding.

How does an LLM generate text?

An LLM reads the input, converts it into tokens, understands the context, and predicts the most likely next token repeatedly until it finishes the response.



Difference between Traditional ML and LLMs.

Traditional ML is usually built for one specific task and often needs labeled data and feature engineering. LLMs are trained on large amount of text data which can understand natural language, and can perform many tasks from the same model using prompts.

What are Tokens?

Tokens are small pieces of text that an LLM processes. A token may be a whole word, part of a word, a punctuation mark, or even a space which depends on the tokenizer.

What is Tokenization?

Tokenization is the process of breaking text into tokens before the model processes it.

What is Context Window?

The context window is the maximum number of tokens an LLM can consider/process at one time. A larger context window lets the model remember more of the conversation or document.

What are Embeddings?

Embeddings are numerical representations of words, sentences, or documents. Words with similar meanings are placed close together in this mathematical space, helping with search and retrieval.

What is Prompt Engineering?

Prompt engineering is the practice of writing clear and detailed instructions for Ai model so an AI model produces better results.

Temperature

Temperature controls randomness of Ai model. Low temperature gives more predictable answers, while high temperature produces more creative responses.

Top-p

Top-p controls how many possible next words the AI can choose from. A lower Top-p gives more focused answers, while a higher Top-p allows more variety and creativity.

Which words am I allowed to pick from?

Max Tokens

Max Tokens sets the maximum length of the LLM model's response.

Difference between GPT, Gemini, Claude, Llama

GPT (OpenAI) is widely used for chat, coding, and reasoning. Gemini (Google) integrates well with Google's ecosystem and supports multiple data types. Claude (Anthropic) is known for strong writing and safety. Llama (Meta) is available in open-weight versions that developers can run and customize according to their requirments.

What is an API? Why do LLMs require API keys?

An API(Application Programming Interface) allows one software application to communicate with another. API keys identify who is making requests, help control access, track usage, and protect the service from misuse.

What are the limitations of LLMs?

LLMs can make mistakes, have outdated knowledge if not connected to current data, may misunderstand unclear prompts, can reflect biases from training data that contain wrong information, and require significant computing resources.

What is Hallucination in LLMs?

Hallucination happens when an LLM confidently produces information that sounds correct as seen by human but is actually false or unsupported.